

ISRO SATELLITE LAUNCH

Complete Mission Analysis

Introduction

The Indian Space Research Organisation (ISRO) is the national space agency of India responsible for designing, developing, and launching satellites and space missions. Established in 1969, ISRO has played a major role in transforming India into one of the leading space-faring nations in the world.

ISRO's satellite launch missions support various sectors including communication, weather forecasting, navigation, agriculture, disaster management, scientific research, and national security. Through continuous innovation and advanced technology development, India has achieved significant milestones in space exploration.

The year 2025 witnessed several important satellite launches by ISRO, strengthening India's capabilities in Earth observation, communication, and scientific research. These missions demonstrated India's commitment towards using space technology for national development and global cooperation.

History of ISRO

The Indian space programme began with the establishment of the Indian National Committee for Space Research (INCOSPAR) in 1962 under the leadership of Dr. Vikram Sarabhai, who is considered the father of the Indian space programme.

In 1969, INCOSPAR was transformed into the Indian Space Research Organisation (ISRO). The organisation focused on developing indigenous satellite technology and launch vehicles to achieve self-reliance in space activities.

Over the years, ISRO developed several successful launch vehicles including:

- Satellite Launch Vehicle (SLV)
- Augmented Satellite Launch Vehicle (ASLV)
- Polar Satellite Launch Vehicle (PSLV)
- Geosynchronous Satellite Launch Vehicle (GSLV)
- Launch Vehicle Mark-III (LVM3)

Important Satellite Missions

1. Earth Observation Satellites

Earth observation satellites help monitor natural resources, climate changes, agriculture activities, forests, oceans, and disaster situations.

Applications include:

- Weather monitoring
- Agricultural planning
- Flood and cyclone prediction
- Environmental studies
- Mapping and surveying

2. Communication Satellites

Communication satellites provide services such as:

- Television broadcasting
- Internet connectivity
- Telephone communication
- Digital education services
- Emergency communication systems

These satellites help connect remote areas and support digital development across India.

3. Navigation Satellites

India developed its own navigation system called Navigation with Indian Constellation (NavIC).

NavIC provides:

- Accurate location services
- Vehicle tracking
- Disaster management support
- Maritime navigation
- Defence applications

Major ISRO Launch Vehicles

Polar Satellite Launch Vehicle (PSLV)

PSLV is one of ISRO's most successful launch vehicles. It is known for reliability and has launched many Indian and foreign satellites.

Features:

- Designed for launching satellites into polar and sun-synchronous orbits
- High success rate
- Used for Earth observation missions

Geosynchronous Satellite Launch Vehicle (GSLV)

GSLV is used to launch heavier satellites into geostationary orbit.

Features:

- Uses advanced cryogenic engine technology
- Supports communication satellite launches
- Improves India's independent launch capability

LVM3

Launch Vehicle Mark-III is ISRO's heavy-lift launch vehicle.

Applications:

- Human space mission support
- Heavy satellite launches
- Deep space mission support

Importance of Satellite Launches

Satellite launches contribute to India's development in many areas:

Agriculture

Satellites provide information about:

- Crop monitoring
- Soil conditions
- Weather patterns
- Water resources

Disaster Management

Space technology helps in:

- Cyclone tracking
- Flood monitoring
- Rescue operations
- Emergency communication

Defence and Security

Satellites support:

- Border monitoring

- Communication systems
- Surveillance capabilities

Education and Healthcare

Satellite communication helps provide:

- Online education
- Telemedicine services
- Digital connectivity

ISRO Achievements

Some major achievements of ISRO include:

- Successful Mars Orbiter Mission (Mangalyaan)
- Chandrayaan lunar exploration missions
- Development of indigenous launch vehicles
- Successful commercial satellite launches
- Advancement of space technology research

Future Missions of ISRO

ISRO continues to work on ambitious missions including:

Gaganyaan Mission

India's first human spaceflight programme aimed at sending Indian astronauts into space.

Chandrayaan Missions

Future lunar exploration missions will focus on advanced scientific research and exploration.

Solar Missions

Missions like Aditya-L1 study the Sun and help scientists understand solar activities.

Conclusion

ISRO's satellite launch programme represents India's technological advancement and scientific achievements. Through continuous research, innovation, and successful missions, ISRO has established India as a major player in the global space sector.

Satellite technology will continue to play a crucial role in communication, agriculture, disaster management, scientific research, and national development. ISRO's future missions will further expand India's presence in space exploration.